

A Secure Dynamic Mix Zone Pseudonym Changing Scheme Based on Traffic Context Prediction

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Abstract—Traffic context plays an important role in supporting automated driving and intelligent transportation systems. Smart vehicles explore surrounding environments by analyzing sensor data and periodically communicating with neighbors and road infrastructures. The context can be well learned in this way to support driving, but the vehicle trajectory can be also easily exposed under eavesdropping attacks. The pseudonym is proposed to hide the real identity of the vehicles. However, the effectiveness of anonymity, the safety of driving, the convenience of implementation and the utilization of resources in previous approaches have not been well-balanced. Therefore, focusing on efficiently replacing pseudonyms with the premise of ensuring driving safety, we propose a secure dynamic silent mix zone pseudonym changing scheme (TLAS) based on the real-time traffic context prediction for urban regions. It naturally takes the area in front of the red traffic light as a silent mix zone, which avoids the driving security issue caused by signal silence. Besides, the area length is dynamically configured according to the traffic context predicted in the last green light cycle, so the anonymous effect can be improved. In addition, considering the resource utilization and accuracy requirement, the adaptive prediction algorithm is applied. We conduct simulation experiments with real-world traffic history using SUMO and OMNET++, the results show that TLAS strategy can indeed achieve a better anonymous effect (reducing standardized traceability rate by 8.2%) with lower driving speed for safety concern.

Index Terms—Trajectory privacy, security, pseudonym changing, traffic prediction.

I. INTRODUCTION

WITH the advance of smart sensors, big data, artificial intelligence and mechanical manufacturing technologies, the connected vehicles and intelligent transportation fields are blooming. The global connected vehicle market is estimated to reach about 166 billion in 2025 [1]. Traffic

context plays an important role in helping vehicles understand the surrounding environment and navigate autonomously [2], [3]. A single vehicle cannot analyze and aware of such complex context by itself, data transmission is necessary to build the full view. So, the flexible vehicle ad-hoc network is widely used. Each vehicle is a node, and it can communicate with all surrounding environments (V2X), including other vehicles, infrastructures, pedestrians, and so on [4], [5]. The decentralized dynamic self-organizing real-time network can support both safety services (such as collaborative driving, accident warning) and entertainment applications like music sharing [6]. According to the standardized development organization regulation [7], vehicles should broadcast current status by 1-10 HZ frequency while driving. The data is called beacon, including vehicle identifier, position, speed, direction, etc. It helps vehicles to obtain environmental context in advance, so the vehicles can cooperate and remind the driver of the potential upcoming accident to improve driving safety. However, the data is wireless broadcast to everyone, an eavesdropping attacker can easily collect these information with a low-cost receiver and get the detailed movement of target vehicles [8]. For example, the attacker can obtain when the driver goes out, when he goes home, where he works, etc. It seriously violates the privacy of the driver, even endangers his/her safety and property. Therefore, it is essential to protect the trajectory privacy of vehicles in intelligent transportation systems.

To solve the privacy problem, pseudonym schemes are widely applied, which leverage different multiple pseudonyms to hide the vehicle identity. The vehicle security standards IEEE1609.2 [9] and ETSI102941-v1.1.1 [10] describe the deployment of a security architecture based on public key infrastructure (PKI), it uses a trusted authority (TA) to distribute encrypted pseudonyms. Specifically, a vehicle will firstly get an initial set of pseudonyms and private keys after the registration. Then, it can change the pseudonyms during the driving and request a new group of pseudonyms after used up. Different identifiers can help ensure the trajectory privacy of the vehicle while satisfying the network requirement of broadcasting the beacon. The key point of the pseudonym scheme is to decide when and where to change the assigned pseudonym, which directly influences the anonymous performance. On one hand, the pseudonym changing group should have as many vehicles as possible; On the other hand, all of the vehicles in the group better change the pseudonym exactly

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at the same time to make it more confounding. Meanwhile, driving safety also needs to be considered as the first priority. Hence, how to safely, efficiently and effectively leverage the pseudonyms to protect vehicle trajectory privacy is still facing huge challenges.

There are a lot of pseudonym schemes that have been proposed recently, which can be categorized into the mix zone based strategies and the mix context based strategies [11]–[13]. Mix zone based strategies are widely used in the urban area. They usually select intersections, parking lots, gas stations and other traffic-intensive locations to change pseudonyms. However, if there are not many vehicles in these fixed locations, the anonymous effect will be greatly reduced. Mix context based strategies arrange the pseudonyms changing according to the surrounding environment, such as traffic density, vehicle speed, tracing probability, etc. All vehicles collaboratively change or exchange pseudonyms at the same time to hide their real identity. Due to the high flexibility, mix context based strategies are commonly applied on the way (like a highway) without road infrastructures. Since beacons are continuously broadcast to guarantee driving safety, the anonymous effect is not always good [14]. Moreover, vehicles complete the whole pseudonym changing process without any infrastructure help, they should have enough resource and energy support.

In this paper, a secure dynamic silent mix zone pseudonym changing scheme TLAS is proposed. To provide a more secure and flexible scheme, we select the mix zone in a precise manner. Instead of taking the whole square intersection as the pseudonym changing location, TLAS sets the four areas in front of the traffic light as the potential silent zone. Since all vehicles slow down and stop when the traffic light is red, driving safety will not be influenced a lot by missing beacons. Besides, the silent zone length is dynamically configured according to the traffic context predicted in the last green light cycle. Specifically, we focus on the traffic flow as the main context. In this way, we can guarantee a certain amount of vehicles in the mix zone and avoid the poor anonymous effect. Moreover, the prediction accuracy influence on the anonymity and safety, as well as the corresponding resource costs are also considered. TLAS selects the most suitable prediction algorithm based on the current conditions. Generally, according to the variable green traffic light time, accuracy and resource requirements, a prediction algorithm is applied to estimate the potential vehicle amount in the next red light cycle. Then, TLAS dynamically sets the mix zone area and the intersection infrastructure broadcasts the silent position information when the traffic light turns red. After vehicles arrive, they stop send beacons and start to change the pseudonym. When the light goes to green, they exit the silent zone with new identities.

Our contributions can be summarized as follows:

- We propose TLAS, a secure dynamic silent mix zone pseudonym changing scheme based on traffic context prediction for the urban region. By leveraging the low-speed nature of vehicles under red light, safety is improved in the precise silent zone even with no beacons. Besides, TLAS dynamically configures the mix zone length based on the real-time predicted traffic context. The flexible approach can better protect trajectory privacy.

- Considering the real-world scenario that traffic light duration various along the time, a prediction model is designed. It adaptively selects a suitable prediction algorithm based on multiple factors, including prediction accuracy, running time, resource consumption (like memory and CPU), and so on.
- Comparing with three traditional pseudonym changing schemes, comprehensive evaluation experiments (including mix zone length, error coefficient, candidate prediction algorithms) are conducted from the aspects of traceability and driving safety. The simulation results show that TLAS can indeed protect trajectory privacy and securely achieve considerable anonymity with low resource utilization.

The rest of the paper is organized as follows. We introduce the related work and limitations of pseudonym schemes in Section II. The system architecture is presented in Section III, including pseudonymous strategy, adversary model and privacy metrics. Section IV illustrates the detailed implementation of TLAS scheme, as well as the traffic flow prediction method. Section V demonstrates the comprehensive evaluation results using SUMO and OMNET++ simulators. Finally, conclusions and future work are listed in Section VI.

II. RELATED WORK

To protect trajectory privacy, pseudonym changing strategies are widely used. Different pseudonyms can help hide vehicles' real identity and break continuous trajectories. The key is to select the proper time and location to replace the pseudonym without affecting vehicle driving safety. Existing pseudonym schemes can be categorized into the mix context based strategies and the mix zone based strategies [11]–[13].

In mix context based strategies, vehicles continuously detect the surrounding environment while driving and coordinate with neighbors to change the pseudonym simultaneously. The concept of mix context is first introduced by Gerlach and Guttler [15], vehicle can initiate pseudonym changing process when the neighbors reach a certain amount. Based on that, many researchers proposed various optimization approaches. For examples, Pan and Li [16] took the status and cooperation intention of neighbors into the consideration, the context condition became more complex. Liao *et al.* [17] added a flag bit to the traditional pseudonym changing protocol and proposed the maximum waiting time mechanism. David Eckhoff *et al.* [18], [19] allowed neighbor vehicles to exchange pseudonyms to enhance privacy. Boualouache and Moussaoui [20] proposed to cooperate part of the vehicles based on the congestion situation detected by monitoring driving speed. These schemes are recommended in areas with inadequate infrastructures, such as highway or suburbs. In addition, since vehicles need continuously broadcast beacons to guarantee the driving safety, the anonymous effect is limited. However, in urban environments, the traffic situation is complex with multiple roads crossover and the facilities are sufficient. Many vehicles gather in hotspots (such as the intersections, parking lot, etc.), it is nature to apply mix zone based pseudonym changing strategy which is more efficient and convenient.

Mix zone based strategies commonly leverage high traffic density locations as the pseudonym changing area, such as the intersections, parking lots, gas stations, etc. When vehicles enter the mix zone, they stop broadcasting any information and become silent. Then, they replace the pseudonym and exit the zone with a new identity. The mix zone based strategies are usually applied in the urban area with the help of various road infrastructures. Depending on the focus of the strategies, they can be classified into the pseudonym changing policy type and the mix zone design type.

For pseudonym changing, Freudiger *et al.* [21] designed the encrypted mix zone CMIX, where the vehicle information is encrypted before transmission. Vehicles can broadcast their encrypted beacon, freely replace pseudonyms in CMIX and exit the area with a new identity. However, CMIX cannot handle the inner-attacker situation. If an opponent's vehicle is also in the mix zone, it can easily decrypt other vehicles' beacon and link the old-new identity relation. Besides, the overhead of encryption and decryption affect vehicle performance and safety. Boualouache *et al.* [22] proposed the S2SI protocol for pseudonym changing, which provides a silent protocol for building mix zone and an exchange protocol for pseudonym replacing. [23] proposes a scheme of pseudonym exchange between the two most similar vehicles. However, the silent of a busy intersection will influence the normal communications of VANET, which may lead to severe traffic accidents. At the same time, there are still many problems with the pseudonym exchange scheme in the timely accountability of accidents.

For mix zone design, Boualouache *et al.* [24], [25] built VLPZ with the help of road infrastructures. Similar to roadside gas stations and toll stations, vehicles gather in the zone and stop broadcast beacon. They stay for a while, change the pseudonym and choose a random VLPZ lane to exit. In reality, it is not possible to construct many VLPZs along the road only for pseudonym changing purpose. Lu *et al.* [26] took public social zones as the mix zone to change pseudonyms, such as parking lots and intersections. A large amount of vehicles change the pseudonyms at the same time would improve their anonymity. Boualouache and Moussaoui [27] selected the zone in front of the red traffic light as the silent mix zone. Vehicles can replace their pseudonyms or exchange pseudonyms with neighbors by leveraging roadside units (RSUs). However, the zone range did not illustrate clearly and the approach cannot guarantee to protect trajectory privacy under low traffic flow. Palanisamy *et al.* [28] proposed MobiMix, a road network based mix zone, which is a non-rectangle area and configured according to the time window. By leveraging the road center and exit area, MobiMix can better hide the timing correlation of vehicle entry and exit. Although the approach did not put much attention on driving safety and pseudonyms changing results, it directs another design direction for mix zone area.

In a summary, existing approaches cannot well-balance driving safety, flexibility of zone configuration and pseudonym utilization. Hence, we propose a secure dynamic silent mix zone pseudonym changing scheme (TLAS) based on the traffic condition for urban regions. The area in front of the red traffic light is taken as the mix zone. By leveraging the low-speed nature, driving safety can be guaranteed. Besides, the zone

length is dynamically configured according to the real-time traffic flow to maintain the anonymous effect.

III. SYSTEM MODEL AND ARCHITECTURE

In this section, we first introduced the area of mix zone and our pseudonym changing policy. Then, the architecture of TLAS is illustrated in detail. Following that, we present the threat model and the anonymous metrics that used in trajectory privacy protection.

A. Pseudonym Changing Policy

The mix zones of traditional methods are commonly fixed in a circle or rectangle like M1 and M2 in Figure 1(a). All the vehicles in the zone should silent, which may influence the driving security of green light roads. Besides, their anonymity effects depend mostly on the real-time traffic flow. To handle these problems, we set the area in front of the red traffic light as a wireless silent mix zone (showed in Figure 1(b)). The zone width covers all lanes in one direction (D1 - D4), and the zone length is dynamically configured based on the real-time traffic flow. Generally, the zone should cover all vehicles that block (or almost block) due to the red traffic light. In each green light cycle, our system monitors the traffic flow and predicts the zone length for the following red light cycle with the help of road infrastructures like RSU. Noted that the zone length at each direction can be different.

When the traffic light is red, TLAS broadcasts the *SilentZoneStart* signal and the *SilentZonePosition* POS_{SM} that calculated based on the zone length (L_{SM}). If vehicles are in (or reach) the silent mix zone, they will stop their beacon and replace the pseudonym. When the traffic light is green, the *SilentZoneEnd* signal is send out. Vehicles recover beacon communication as long as they leave the mix zone or receive the *SilentZoneEnd* signal. TLAS calculates the actual amount of vehicles and reset the traffic flow *MonitoringPosition* POS_M (and *MonitoringLength* L_M) for the next cycle.

B. Architecture

System architecture are shown in Figure 2, TLAS system includes a cloud layer and an edge layer. Trusted Authorities (TA) commonly run in the cloud servers and belong to government departments. It is composed of registration module and pseudonym generating module. The registration module validates the identity of vehicles, logs the communication records, and distributes the pseudonyms produced by the pseudonym generating module to corresponding vehicles. There are three roles in the edge layer: vehicles, road infrastructures like RSU, and tracking devices like eavesdropper from adversaries. Vehicles carry a data transmission module to communicate with surrounding devices and a pseudonym module which is responsible for its own pseudonym management. Road infrastructure (e.g. RSU, traffic lights, etc.) contains data transmission module, pseudonym distribution module and mix zone module. The pseudonym distribution module performs the local validation of TA and vehicles, then it helps transfer the pseudonyms between them.

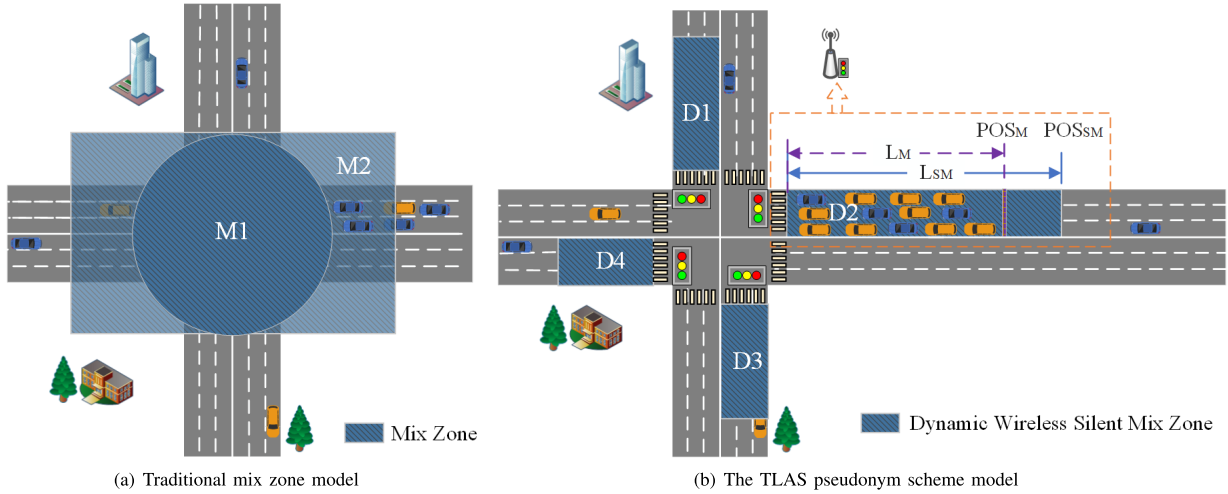


Fig. 1. Comparison between TLAS model and traditional mix zone.

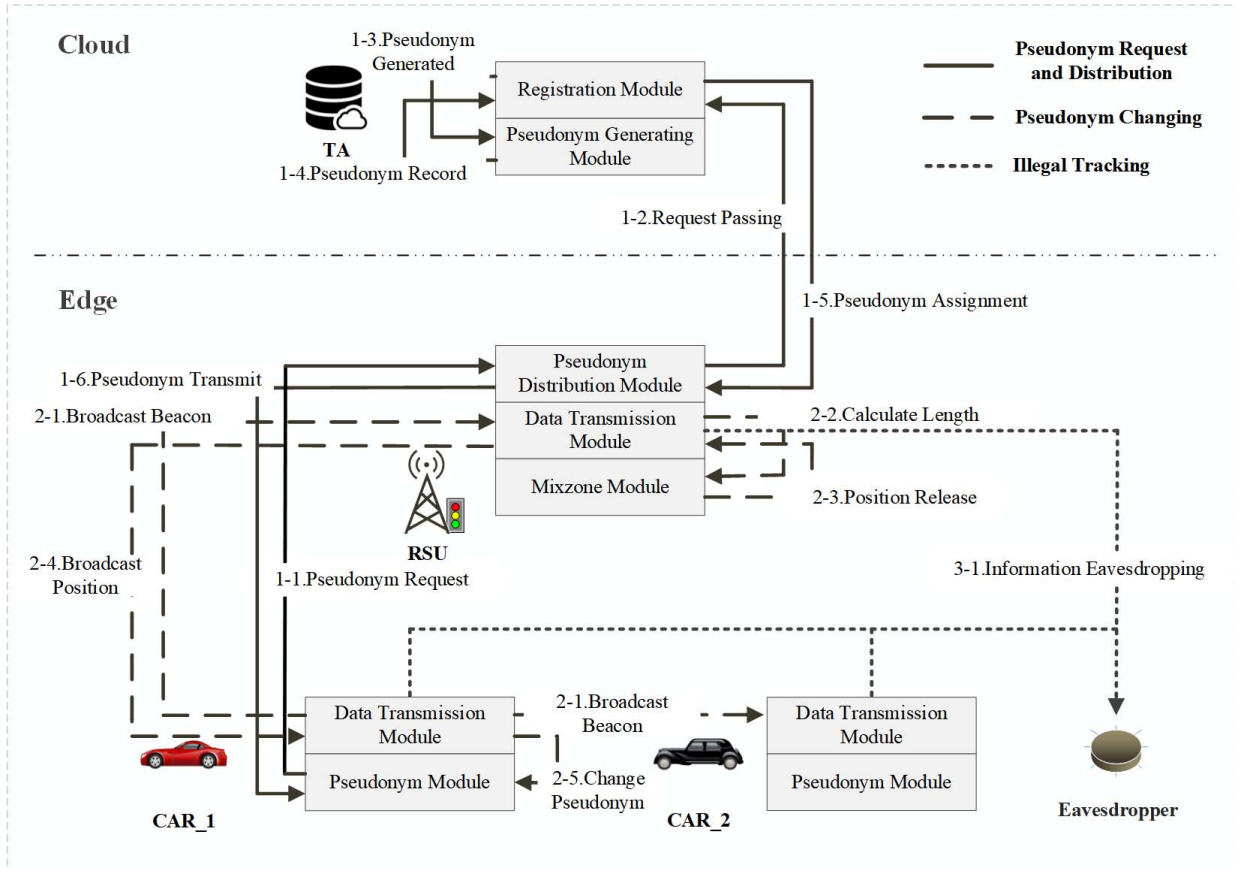


Fig. 2. System architecture.

The mix zone module responsible for calculating the mix zone length in each traffic light cycle by leveraging multiple traffic flow prediction algorithms.

1) *Pseudonym Request and Distribution*: If a vehicle first joins the network or its pseudonyms are almost used up, the vehicle will initial a pseudonym request to the nearest RSU (1-1.Pseudonym Request). After validate the identity of the vehicle, the pseudonym distribution module of RSU passes the request to the TA (1-2.Request Passing). TA receives the request, logs the information, generates a group of

pseudonyms (1-3.Pseudonym Generated) and sends back to RSU (1-4.Pseudonym Record, 1-5.Pseudonym Assignment). Noted that, the RSU may be different depends on the vehicle's current location. Finally, the new pseudonyms are distributed to the vehicle (1-6.Pseudonym Transmit).

2) *Pseudonym Changing*: To guarantee the driving safety, vehicles periodically (at a frequency of 1-10HZ) broadcast their status information, including vehicle identifier (pseudonym), position, speed, direction, system status, etc. Except neighbor vehicles, RSU also gets the information

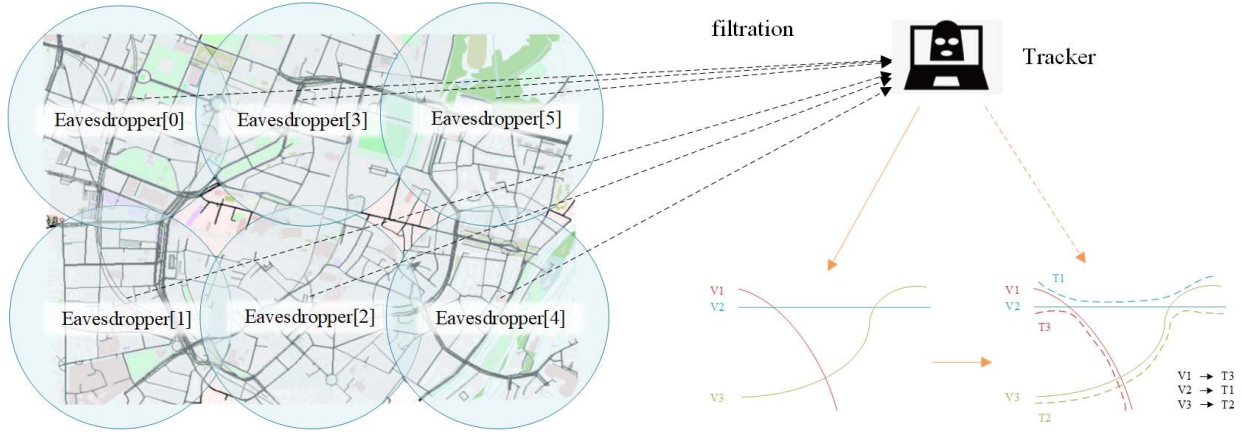


Fig. 3. Global passive adversary threat model.

(2-1.Broadcast Beacon). After collecting the data for a green light cycle, the mix zone module of RSU can predict the traffic flow and calculate the zone length with the road information (2-2.Calculate Length, 2-3.Position Release). When the traffic light turns red, the *SilentZonePositon* POS_{SM} and silent signal *SilentZoneStart* $\{I_{SM}=1\}$ is broadcast to every vehicle in the area (2-4.Broadcast Position). As long as a vehicle receives the information, its pseudonym module will check the pseudonyms it carries and replace the old pseudonym (2-5.Change Pseudonym). Hence, the vehicle will get back on-line to the network with a new identity and continuous drive with beacon to the next intersection after leaving the area or receiving the *SilentZoneEnd* $\{I_{SM}=0\}$ signal.

3) *Illegal Tracking*: Although the broadcast of vehicle beacon helps the driving, it also provides illegal tracking opportunities for adversaries. Cheap data collecting sensors can be widely-spread along the road to monitor vehicle communication data. In TLAS system, We consider a global passive adversary threat model. Adversaries collect the broadcast data through eavesdropper (3-1.Information Eavesdropping), and then analyze the data to reconstruct the trajectory of target vehicles.

C. Threat Model

For the trajectory attack, the adversary aims to identify and track the target vehicle no matter what pseudonym it uses. Normally, the adversary puts sensors along the road and eavesdrops the broadcast information of vehicles and RSUs. After analyzing the collected data, the adversary may crack the pseudonym and obtain the driver's trajectory information like driving route and parking location [29], [30].

As Figure 3 presents, we consider a wide-spread global passive opponent threat model [14] in our system. Since the cost of eavesdropping sensors is low, the adversary can deploy as many as possible to cover the whole urban map. The eavesdropping sensors collect the broadcasting wireless data in the area and send them to the tracker. After filtering the duplicate data, the adversary leverages NNPDA (Nearest Neighbor Probabilistic Data Association) multi-target tracking algorithm [31] to crack pseudonyms and reconstruct the vehicle trajectory. For example, the algorithm reconstructs three vehicle tracks V1,

V2 and V3 (in Figure 3) into the tracking fragments T1, T2 and T3 by calculating the allocation probability matrix. The main objective is to analyze the position correlation before and after vehicles entering the mix zone and identify the target vehicle. The similarity between the reconstructed trajectory and the real path reflects the performance of the adversary. A "almost same" reconstructed trajectory will leads to severe privacy leak of drivers. In the paper, we will use the global passive adversary threat model to simulate the behavior of adversaries and evaluate how TLAS works in such situations.

D. Metrics

The ultimate goal of TLAS is to protect the trajectory privacy of vehicles and guarantee the driving safety. Hence, we use several related metrics to quantitatively evaluate scheme performance.

Privacy metrics present the effectiveness of pseudonym scheme in terms of trajectory privacy protection under threat. There are a lot metrics used previously, such as the size of anonymous set, anonymous entropy, traceability rate, standardized traceability rate, maximum tracking time, and so on [11]. Most metrics reflect the spatial position and speed variation of vehicles or the vehicle distribution, which can be further used to evaluate the performance of pseudonym changing schemes.

Although the anonymous set is easy to calculate, we do not know the information that the adversary obtained in advance. Maximum tracking time pays less attention on the validation of tracking results. Besides, the road distribution greatly influences the anonymous entropy, which is difficult to quantify. Hence, we take the traceability rate and the normalized traceability rate proposed by [32] as the main privacy metrics. As shown in Formula (1) and (2) and below.

The traceability rate:

$$\Pi = \frac{1}{N} \sum_{v \in V} \lambda_v \times 100, \quad \lambda_v = \begin{cases} 1 & \frac{\tau_v}{L(v)} \geq 0.90 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The normalized traceability rate (2), shown at the bottom of the next page. Where v represents the vehicle, $L(v)$ is the lifetime of v , and τ_v represents the traceable time of v . N is the total number of tracks contained in the data set. $psd(v, k_0)$

and $psd(v, k_0 + L(v))$ are the pseudonyms for the first and last time interval of the vehicle v 's life cycle, respectively.

In addition, we take the average speed of nearby vehicles during pseudonym changing into the consideration in Formula (3). It illustrates the overall speed condition of the area, and it is also an important factor for the safety, especially driving in the silence zone.

The average speed of nearby vehicles:

$$\overline{SV} = \frac{1}{N} \cdot \sum_{v=1}^N \left[\frac{1}{k \cdot L(v)} \cdot \sum_0^{psd(v, k_0 + L(v))} \sum_{i=1}^k \tilde{v}(i) \right] \quad (3)$$

where k represents all the neighbor vehicles of the target vehicle v . If the distance of the vehicle i and the target vehicle v is less than the pre-defined distance R , then we treat i as a neighbor and its instantaneous speed $\tilde{v}(i)$ will be used.

IV. TLAS IMPLEMENTATION

In this section, we present the detailed implementation of TLAS scheme. Considering the real-time traffic flow, TLAS dynamically configures the mix zone to guarantee the anonymous effect. Besides, as an important component, traffic flow prediction is also clearly explained.

A. TLAS Scheme

Algorithm 1 demonstrates the main steps of TLAS scheme. Generally, the mix zone is the area in front of the traffic light, while the zone length is dynamically configured according to the real-time traffic situation. The vehicles initial pseudonym changing operation when they satisfy the following two conditions: 1) received the silent signal $I_{SM}=1$, and 2) their current location is in the mix zone. The silent signal is set based on traffic light status, the mix zone only works when the light is red to guarantee the driving safety. The traffic flow is predicted by calculating the amount of new vehicles passed the monitoring site.

Specifically, there are three phrases in the TLAS scheme: traffic data collection phrase, mix zone configuration phrase and pseudonym changing phrase.

1) *Traffic Data Collection Phrase*: When the traffic light turns green, the mix zone silent signal I_{SM} is set to 0. RSU collects the beacon data from all vehicles and estimate the real block length due to the red traffic light. The length will be used to set the monitoring position. Then, all new incoming vehicles that pass the monitoring position are counted and the data is feed to the traffic flow prediction sub-module.

2) *Mix Zone Configuration Phrase*: Since TLAS takes the red traffic light period as the mix zone valid time, the dynamic zone length should be calculated before that. If the traffic light is yellow, we will leverage it to execute the prediction algorithm, otherwise, we take the first three seconds of the red light to do the work. With the input of the time limit and

vehicle amount information, the traffic flow prediction sub-module calculates the amount of vehicles may appear in the following red light cycle. Combining with the intersection lane distribution, the optimal *MixZoneLength* L_{SM} is configured.

3) *Pseudonym Changing Phrase*: When the traffic light turns red, the *SilentZoneStart* ($I_{SM} = 1$) signal and the *SilentZonePosition* POS_{SM} (absolute coordinate calculated by *MixZoneLength* L_{SM} and intersection location) are broadcast to all vehicles through RSU. The vehicles in the mix zone will stop sending beacons and start replace their pseudonym. When they leave the mix zone or rEnd *SilentZoneEnd* $I_{SM} = 0$ signal, they will restart the communication with a new identity.

Algorithm 1 TLAS Algorithm

Input: Map status *map*, The current time t .

Output: Latest TLAS location information POS_{SM} ,
Vehicle silence status $status[vehicle]$.

```

1:  $I_{SM}, POS_{SM} = \text{getRSUBeacon}(t)$ 
2: for all vehicles  $\in$  map do
3:   if  $\text{getPos}(vehicle)$  in  $POS_{SM}$  then
4:     if  $I_{SM} == 0$  then
5:       if greenLight in start then
6:          $POS_{SM} = \text{setDetective}()$ 
7:       end if
8:        $status[vehicle] = \text{nosilent}$ 
9:        $N = \text{setRefreshClock}(POS_{SM})$ 
10:    else
11:      if redLight in start then
12:         $totalVehicle = \text{setPredictModel}(N_{vehicle},$ 
13:           $greenLight.time())$ 
14:         $POS_{SM} = \text{setNewLength}()$ 
15:         $\text{setRefreshBeacon}()$ 
16:      end if
17:       $status[vehicle] = \text{silent}$ 
18:       $\text{setSilent}(vehicle)$ 
19:    end if
20:  end for
```

B. Traffic Prediction

Traffic flow prediction plays an important role in TLAS scheme, it is the basis for dynamic mix zone length calculation and directly influences the vehicle anonymous performance. According to the vehicle flow monitored in the green light cycle, we want to estimate the potential number of vehicles in the following red light cycle. When the vehicles in the mix zone reach a certain amount, the adversary cannot identify the target after they change the pseudonyms. Besides, multiple factors are taken into the consideration, including prediction

$$\Pi_n = \frac{1}{N} \sum_{v \in V} \lambda_v^{\text{norm}} \times 100, \lambda_v^{\text{norm}} = \begin{cases} 1 & \frac{\tau_v}{L(v)} \geq 0.90 \wedge psd(v, k_0) \neq psd(v, k_0 + L(v)) \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

algorithm time complexity, green/red traffic light time proportion, prediction accuracy, resource consumption, and so on. A suitable prediction method will be applied to get the optimal performance.

1) *Traffic Monitoring*: To calculate the number of new vehicles join the intersection/road during the green traffic light, we set a monitoring baseline to distinguish the existing vehicles and incoming vehicles. When the traffic light turns green, the silent mix zone is invalid and all vehicles are sending beacons. Based on the received vehicle location data at this moment, RSU knows exactly how many vehicles are blocked. The block length is taken as the monitoring baseline. Then, we count all vehicles that pass the baseline as the new vehicles, and the data will be feed to the prediction algorithm. Specifically, if the two consecutive locations (obtained from the beacon) of the vehicle is before and after the baseline, it will be treated as the new vehicle. The reason that we set the monitoring baseline to block length instead of a fixed position is to achieve more accurate prediction results. Take the pedestrian line as an example, if it is the monitoring baseline, then the blocked vehicles will lead to a traffic peak in the first few seconds of green light. Since most prediction algorithms are time-based, it is difficult to train an accurate prediction model from the “typical” vehicle flow (a dense in the beginning and sparse later). Depending on the relative location of block length and the monitoring baseline, the lasting time of the traffic peak may different. Therefore, it is necessary to set the baseline as the actual block length to eliminate the man-made traffic peak.

Vehicle speed and adjacent distance are both considered to estimate the block length. When the traffic light turns green, all vehicles send beacons simultaneously and their location data is collected by RSU. Starting from the location of traffic light, each vehicle is examined. If the following two conditions are satisfied, the vehicle is counted as block vehicle: 1) the speed of vehicle is lower than the congestion threshold v_c ; and 2) the adjacent distance to its neighbor is less than the *Maximum Critical Distance* $L\tau$. For each lane, we take the location of the farthest block vehicle as the block length of the lane Q_Lane_i . Figure 4 shows an example of a monitoring baseline setting to calculate the actual queue length of a blocked traffic flow in front of a red traffic light. In lane la , the position of the vehicle v is recorded as $Lane\{la\}_POX_{x_v}$. If the distance of two vehicles is larger than the *Maximum Critical Distance* $L\tau$, then they are not neighbors. Hence, the length of first lane in Figure 4 is Q_Lane1 , and the black vehicle is ignored. After all lanes are calculated, the maximum length (Q_Lane) is used to set the monitoring baseline. Noted that the congestion threshold is commonly configured to 5km/h based on GA 115-1995 “Road Traffic Congestion Degree and Evaluation Method” regulation [33].

2) *Candidate Prediction Algorithms*: Considering the fast generating speed and strong time correlation of traffic data, TLAS selects the following four prediction algorithms to estimate the traffic flow.

a) *Simple average algorithm*: It takes the arithmetic average value of the last period data as the predicted value [34]. This algorithm can quickly obtain good and stable prediction

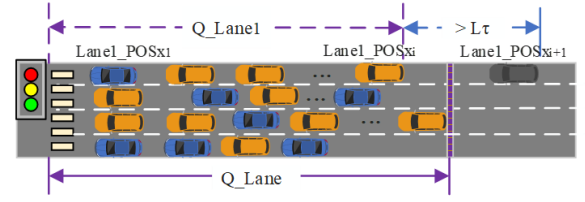


Fig. 4. Monitoring Baseline Setting.

results for datasets with no obvious trend. The general term is shown in Formula (4) below.

$$Pre_{avg}((x_1, x_2, \dots, x_t), n) = \frac{1}{t} \sum_{i=1}^t x_i \quad (4)$$

$((x_1, x_2, \dots, x_t))$ represents the time series data of latest t time steps, and n represents that we want to predict the following n time steps data $((x_{t+1}, x_{t+2}, \dots, x_{t+n}))$.

b) *Auto-arima algorithm*: It is the parameter optimal adaptive algorithm of the commonly used differential autoregressive moving average model Arima (p,d,q) [35]. This model converts the non-stationary time series data into stationary time series data. Only three data types are used in regression: the lag value of the dependent variable, the current value and lag value of the random error variable. This algorithm has the automatic tuning ability, which helps it obtain better prediction results compare to traditional statistical time series prediction algorithms. The general term is shown in Formula (5) below.

$$Pre_{arima}((x_1, x_2, \dots, x_t), n) = \frac{(1 + \sum_{i=1}^q \theta_i L^i) \varepsilon_{t+k}}{(1 - L)^d (1 - \sum_{i=1}^p \phi_i L^i)}, \quad t+1 \leq k \leq n \quad (5)$$

p is the number of auto-regressive parameters, q is the number of moving average parameters, d is the difference order that makes it a stationary series, L is the hysteresis operator, where $L^i = x_{t+1-i}/x_{t+1}$.

c) *Prophet algorithm*: It is an open source algorithm firstly proposed by FaceBook, which supports intelligent prediction by allowing users to configure the time they need, such as the cycle or holidays [36]. This algorithm is preferred to use in long-term prediction with weak trend. Hence, the accuracy will be decreased if the data size is too small. With more training data, it can quickly converge. The general term is shown in Formula (6) below.

$$Pre_{prophet}((x_1, x_2, \dots, x_t), n) = \frac{C}{1 + e^{-k(t+1-b)}} + \varepsilon_{t+k}, \quad t+1 \leq k \leq n \quad (6)$$

$\frac{C}{1 + e^{-k(t+1-b)}}$ represents the logistic regression model of nonlinear growth, where C is the carrying capacity, k is the growth rate, b is the offset, and ε_{t+k} is the noise which indicates the fluctuation.

d) *LSTM algorithm*: LSTM is short for long and short time memory network, it is an improved algorithm of recursive neural network (RNN). The next sliding window can be predicted according to memory statistics of the results of each model, and the final model can be obtained by recursively

training the length of the sliding window. A good result effect for long training and short prediction can be achieved with large number of iterations. The general term is shown in Formula (7) below.

$$Pre_{lstm}((x_1, x_2, \dots, x_t), n) = o_{t+n} \cdot \tanh(C_{t+n}), \\ t + 1 \leq k \leq n \quad (7)$$

o_{t+n} and C_{t+n} represents the predicted input and forget gate, respectively.

3) *Prediction Model*: The output of the prediction model is the vehicles' amount in the next red traffic light cycle. The accuracy of the prediction directly influences the pseudonym anonymous effect. Besides, the prediction model's calculation time is limited. Although we may get more accurate value with complex algorithms, their required time also increased. Since we leverage the green traffic light time to collect data and apply the predicted result in the following red traffic light, the calculation time should strictly follow the traffic light time distribution. Moreover, with the widespread deploying of road infrastructures in urban intersections, RSUs are main components of the modern intelligent transportation system. Trajectory protection, as a supplementary function for personal drivers, should not occupy much public resources. Hence, the resource cost in traffic prediction need to be considered. To well-balance these crucial factors, our prediction model aims at leveraging the minimum resources to accurate predict the traffic flow information on time. Four candidate prediction algorithms are applied to establish the optimal prediction model in different scenarios.

a) *Objective function*: For resource consumption, we take the CPU and memory usage as an example, other resources can also be added in the extended version. The objective function $MinQ$, representing the minimization of resource consumption, is defined in Formula (8):

$$MinQ = \gamma_1 * Pre_{CPU} + \gamma_2 * Pre_{Memory} \quad (8)$$

Pre_{CPU} and Pre_{Memory} represent the CPU and memory consumption of the prediction algorithm, respectively. Their corresponding weights are denoted as γ_1 and γ_2 , where $\gamma_1 + \gamma_2 = 1$. Noted that other resources can also be considered, while the summation of the weights should always be 1.

b) *Constraints*: We have the following two constraints in the prediction model:

- **Calculation Time**: The dynamic mix zone length depends on the predicted traffic flow in the next red traffic light cycle. The prediction raw data is collected in the green light cycle, so the calculation should be finished within the time period of the yellow traffic light or the end of green traffic light (if there is no yellow light time exists).
- **Prediction Accuracy**: Accurate traffic flow prediction can lead to a more precise mix zone length, which further decreases the influence to the anonymous effect and enhance the stability. Considering there are multiple lanes in most intersections in real world scenarios, we use an error coefficient w to restrict the prediction accuracy. The smaller the error coefficient is, the higher the required prediction accuracy will be.

TABLE I
SIMULATION PARAMETERS

Category	Parameter	Default Value
Sumo	Scene Information	Munich City,
		Arrival Rate (1,1.43,2) Veh/s
		Single Signal Intersection, Single Junction (20,40,60) Veh
	Detector Length	Bologna City,
		Real Traffic Flow in 8:00-9:00
		(40,50,60,...,140) m Automatic Adjust, min 40m
Veins	Beacon Rate	1HZ
	Transmission Power	20mW
Tracker	Eavesdropper Range	300m
	Eavesdropper Position	Full Coverage and Overlap 30m
	Track Interval	1s

Therefore, the following prediction model constraint equation can be obtained in Formula (9):

$$\text{s.t.} \begin{cases} t(Pre) \leq t_{yellow} \\ accN(Pre) \leq n_{lane} * w(L_M) \\ \gamma_1 + \gamma_2 = 1 \\ Pre \in \{Pre_{avg}, Pre_{arima}, Pre_{prophet}, Pre_{lstm}\} \end{cases} \quad (9)$$

$t(Pre)$ represents the execution time of the prediction algorithm, and t_{yellow} represents the time of the yellow traffic light, unit is second. $acc(Pre)$ represents the prediction accuracy, n_{lane} represents the number of lanes in front of the traffic light in one direction, and w is the error coefficient. Pre_{avg} , Pre_{arima} , $Pre_{prophet}$, Pre_{LSTM} represent the four candidate prediction algorithms introduced previously.

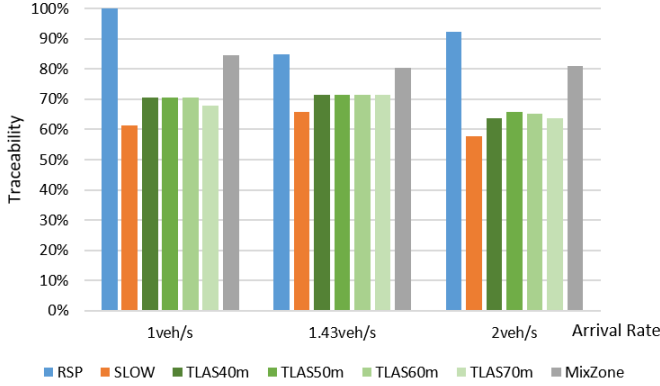
V. EVALUATION

In this section, we present the comprehensive evaluations for our TLAS scheme. Firstly, the experimental environment are demonstrated, SUMO traffic simulator and OMNET++ network simulator are applied. Then, the main design components are evaluated, including mix zone length, error coefficient and candidate prediction algorithms. After analyzing their influence to the anonymous effect, the performance of TLAS scheme with real world traffic flow in Bologna City [37] are demonstrated by comparing with three traditional silent anonymity schemes.

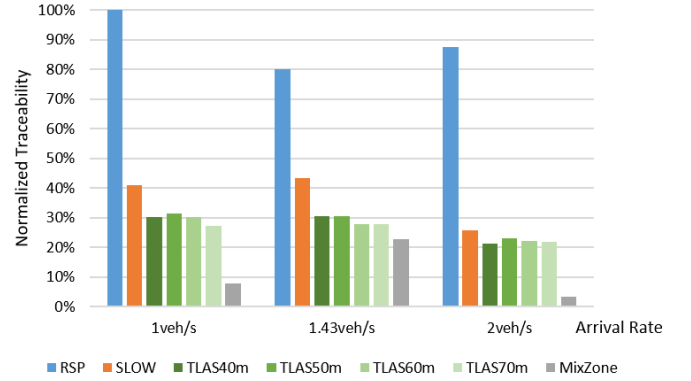
A. Experimental Environment

The experiments are run in Ubuntu 16.04 system with the SUMO traffic simulator, OMNET++ network simulator and its two-way coupling simulation framework Veins. Two types of maps in the experiments: 1) real world map and traffic flow are used to evaluate the TLAS scheme performance 2) single traffic light intersection abstracted from the real world map is used to evaluate the influence of each component. The specific scenario settings and simulation configurations are shown in Table I.

There are three scenes used in the evaluation: 1) the Munich city map with three vehicle arrival rate [14] is to demonstrate



(a) Comparison of traceability



(b) Comparison of normalized traceability

Fig. 5. The anonymity of each anonymous scheme at different arrival rates.

the feasibility and the anonymous of the dynamic mix zone length. The map and traffic flow are from the original PREXT method for better comparison; 2) the single intersection map with different traffic flow to evaluate the prediction algorithms. To eliminate the influence of other factors, we extract a single intersection from the map and run experiments with different traffic flows; and 3) the real traffic data in bologna city map [38] for the performance comparison of TLAS and traditional schemes. It is a real city map with traffic history from 8:00 to 9:00 on the official website of SUMO. Due to the experimental environments, 50 vehicles were selected to evaluate the effects of the pseudonym scheme in real city scenes.

The typical pseudonym schemes used in the comparison are RSP, SLOW and Mixzone. RSP scheme allows the vehicle to change the pseudonym in every fixed time cycle, and all vehicles will silent for a random period of time (within a preset range, like 3-13 s) before changing [39]. SLOW is a slow speed scheme [40], vehicles only broadcast beacons when their speed is higher than the preset threshold V_{top} . If a vehicle does not send beacons for consecutive T_{top} seconds, which means its speed is low for a while, it will change the pseudonym. Mixzone is a classic mix zone based approach [41], vehicles can replace their pseudonym if they are in the mix zone. Normally, the adversary cannot detect the broadcast data in the mix zone, and the zone is usually set on the heavy traffic intersections.

B. Mix Zone Length

The main feature of TLAS is the mix zone configuration, the scheme can dynamically set the mix zone length according to the current traffic flow situation. Theoretically, large mix zone contains more vehicles, which helps improve the anonymous effect. However, considering all the vehicles are silent (stop sending any beacon) in the mix zone, the possibility of security emergency also higher with the increasing amount of vehicles. Hence, we evaluate the influence of various mix zone length, the comparison results with three traditional silent schemes are also presented.

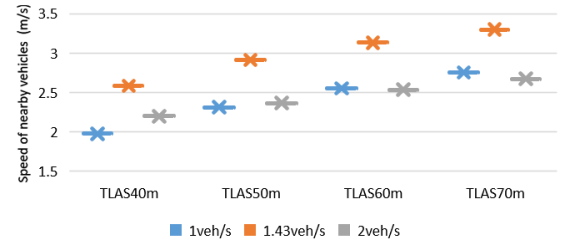
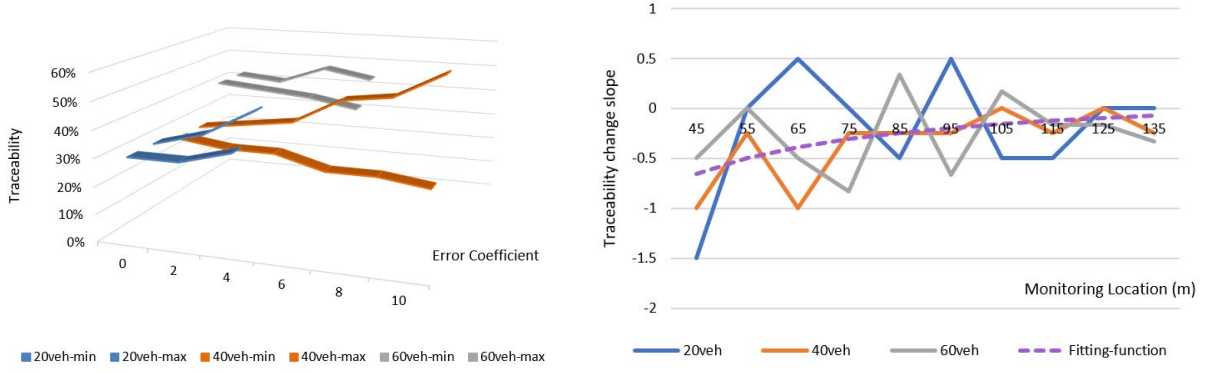


Fig. 6. The driving safety of TLAS at different arrival rates.

In the experiments, the Munich downtown map scene [14] is used to make it more close to real scenario. Since mix zone length relates to traffic flow, we set the vehicle arrival rate to 1, 1.43 and 2 vehicles per second (based on [14]) to simulate different traffic situations. The vehicle trajectory on the map is randomly generated by SUMO. The whole simulation time is 120 seconds, where new generated vehicles can join the traffic in the first 100 seconds. For mix zone length, we select 40m, 50m, 60m and 70m as the value. Based on the national standard of road traffic sign GB5768.3, the length of guidance lane line should be determined according to intersection geometry and traffic management needs, generally no less than 30m. Hence, the minimum silent mix zone length is set to 40m to ensure that vehicles can complete lane change in the silent area to enhance identity confusion. From the aspect of adversary, the eavesdropper performs real-time monitoring in the mode of full coverage, and the NNPD algorithm is applied to splicing the trajectory. In addition, several typical silent anonymous schemes, such as RSP, SLOW and Mixzone, are selected as comparison schemes, and the experimental settings are follow [14] to be consistence.

The traceability indicates the anonymity performance of the whole map, including vehicles in the silent mix zone and normal vehicles which broadcast beacons in other locations, and presents the applicability of the tested anonymous scheme. The normalized traceability, focusing on the silenced vehicles only, evaluates the identity confusion degree of the anonymous schemes. As Figure 5(a) demonstrated, the results of the



(a) Comparison of traceability with error coefficient (the z-axis is used to distinguish the overlapped results, no specific meaning.)

(b) The traceability rate under different monitoring position.

Fig. 7. Influence of error coefficient and TLAS length on the anonymity.

TLAS, SLOW and Mixzone solution all show less than 80% traceability, which indicates that the interruption of unified space-time information (beacon silent) has a great impact on trajectory recovery. With the increase of the mix zone length, the traceability and normalized traceability of TLAS tend to decrease. That is, more vehicles passing through the silent area will lead to lower traceable probability, and the tracking difficulty increases.

In addition, we apply the average speed of nearby vehicles in the monitoring area as the driving security indicator. Since the driving speed is mostly in the range of 20km/h-40km/h in the city and the safe distance between vehicles should be no less than 10m-30m, we set the neighbor radius R to 20m. As Figure 6 demonstrated, the average speed increases linearly with the increasing of the mix zone length.

The goal of TLAS is to ensure both anonymous effect and driving safety. Hence, the optimal silent mix zone length should be carefully configured based on the real environment context.

C. Error Coefficient

The mix zone length in TLAS depends on the predicted traffic flow, which is calculated using candidate prediction algorithms. When we leverage a predicted value, the error is inevitable. Hence, TLAS applies an error coefficient to restrict the accuracy of the predicted traffic flow.

The goal is to analyze the relationship between the prediction accuracy (error coefficient) and the anonymous effect. In this experiment, we take a simple single intersection as the testing map to minimize the impact from the map itself. Similarly, various traffic flow and mix zone length are used to evaluate the error coefficient. The simulation time is over two traffic light cycles, the passed by vehicle numbers are 20, 40 and 60. The range of monitoring position (/ mix zone length) is 40m-140m, where the length step is 10m.

As shown in Figure 7(a), it represents the relationship of tractability and error coefficient under different traffic flows. For example, when there are 40 vehicles in our experiments, the block length of the vehicle queue at the red traffic light is about 90m. When the error coefficient of the prediction

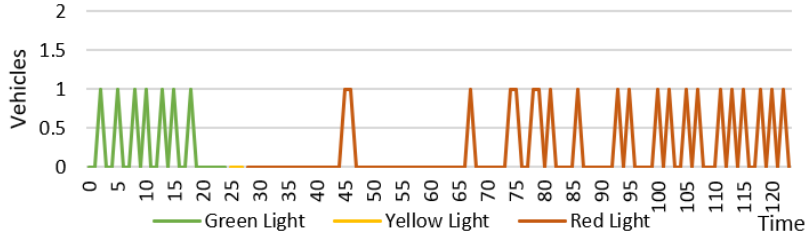
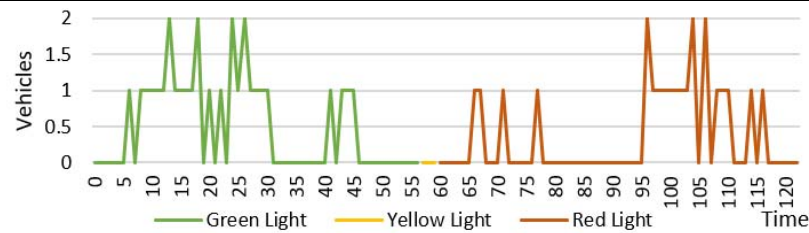
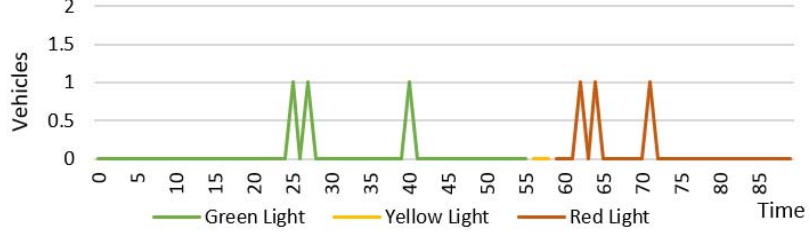
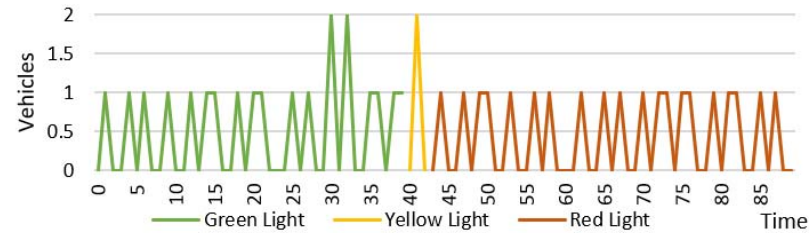
algorithm is 2, the length may be predicted to be 80m or 100m. We took their traceability rate as the result of 40veh-min and 40veh-max respectively, to better demonstrate the traceability fluctuation. Since the error coefficient indicates the prediction error, it leads to different mixed zone length, which further affects the variation of traceability. With the increase of the error coefficient, the fluctuation of traceability also increases.

To evaluate the influence of the baseline input (monitoring position), Figure 7(b) demonstrates the change of the traceable rate slope with the increase of the monitoring position. It shows that the traceable slope gradually decreasing and always less than 0 with the monitoring position increasing, and the traffic flow has negligible impact. When the monitoring position is within the actual block length, the system will count part of the blocked vehicles as the new arriving vehicles. It gives the wrong baseline data to the prediction model. Hence, the situation will unlikely happen when the monitoring position increases. The predicted mix zone length will be more accurate, which leads to a stable traceable rate. The fitting curve of the three traffic flow's average traceable rate slope is the dash line in the Figure 7(b). When the influence of the monitoring position on the traceable rate is large, the allowed error coefficient should be less to guarantee relatively stable traceability. Since the allowable error coefficient is inversely correlated with the traceable slope, assuming the maximum allowable error coefficient of each lane is between 2-6 vehicles, the error coefficient w can be obtained as $w = \frac{270}{180-Lm}$ based on the regression result.

D. Prediction Algorithms

Prediction algorithms leverage the monitoring vehicle data in green traffic light to predict the traffic flow in the following red traffic light. Then, combining the intersection lane information, TLAS can dynamically set the optimal mix zone for vehicle pseudonym changing. Based on the design of TLAS, there are three factors that influence the prediction algorithm selection: algorithm running time, prediction accuracy and resource consumption. Hence, we analyze the four candidate prediction algorithms (Simple average algorithm, Auto-arima

TABLE II
THE NEW VEHICLE DATA PER SECOND

Position	Lane Number	New Vehicle Data Per Second	Traffic Light (Green:Yellow:Red)
Lane: b8	4		24s: 3s: 96s
Lane: b1[1]	3		57s: 3s: 63s
Lane: b35[0]	1		56s: 3s: 31s
Lane: a43[1]	3		40s: 3s: 47s

algorithm, Prophet algorithm, and LSTM algorithm) from the three aspects for helping the algorithm selection.

To simulate the real world scenarios, open source bologna map and its corresponding traffic flow provided by SUMO are used in the experiments. The data set includes the road information, traffic flow of buses and pedestrians during the rush hour in Bologna (8:00-9:00 am). To better compare the prediction performance of different algorithms, we select four distinguish traffic flow in one traffic light cycle as our traffic data, the detailed information are shown in Table II. The proportion of green light time is configured differently to evaluate multiple situations. The traffic data monitoring in green light part is taken as training set, and the vehicle data in the following red light part is treated as test set. Each experiment run five times, the average data is taken as the final result.

After run the experiments using the four traffic flows, we found that the algorithms' performance present the same results and the impact of traffic flow is negligible. Hence, we take the average data of four traffic flows as the presenting

data in Figure 8. As Figure 8(a) shows, simple average, auto-arima and prophet can finish the prediction within three seconds. Considering the yellow light in real world is also around three seconds, we modified the iteration parameter of LSTM to make sure its running time is within the threshold. For error coefficient, Figure 8(b) demonstrates that the result of the simple average and auto-arima is relatively low and stable. Prophet and LSTM converge gradually with the increasing of training set data, and the convergence speed of LSTM is faster. According to the data showed in Figure 8(c), the CPU utilization of prophet and LSTM increases linearly with the training data size, while the trend of simple average and auto-arima are generally stable. For the four prediction algorithms, the CPU utilization of auto-arima is the highest. As Figure 8(d) illustrates, the memory usage of all four algorithms are stable, where the occupancy satisfy the following relation: Simple average < Auto-arima < Prophet < LSTM.

In summary, according to the experiments of four different traffic intersections, we have basically the same prediction

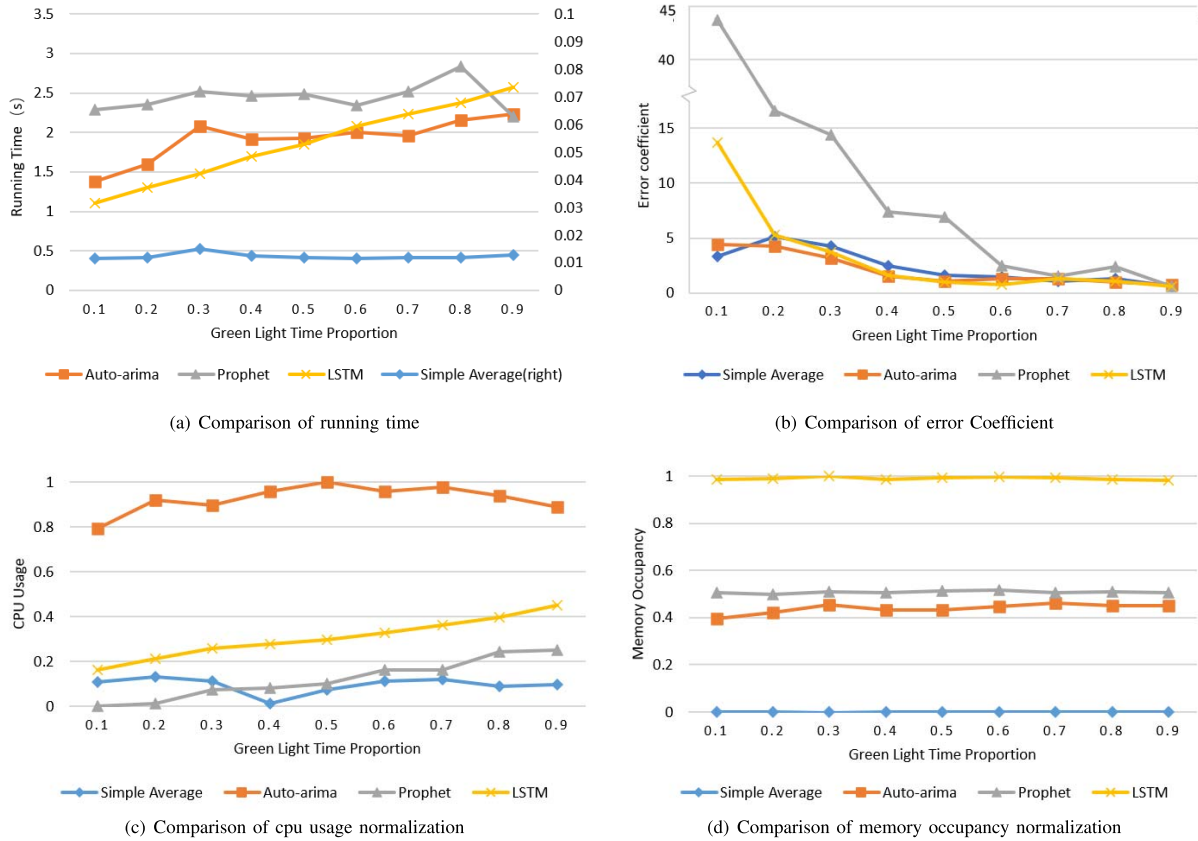


Fig. 8. Performance Comparison of Four Prediction Algorithms.

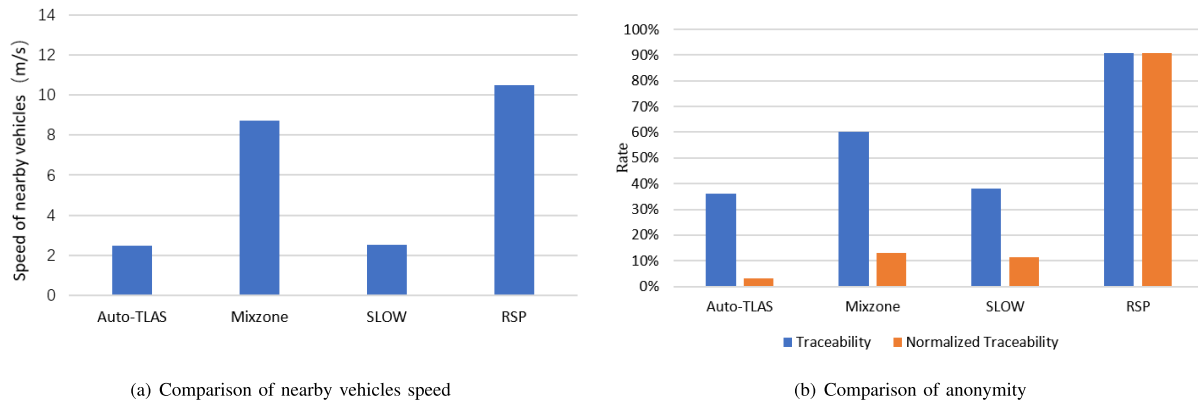


Fig. 9. TLAS performance comparison with the basic pseudonym scheme.

results and performance. TLAS will dynamically select and configure the optimal prediction algorithm based on the objective function and constraints.

E. TLAS Performance

To evaluate the TLAS performance in real world scenarios, the Bologna map is applied and the traffic data is from 8:00 to 9:00 on weekdays [38]. The experiment data set includes 50 vehicles' trajectory in 120 seconds. Comparing with three typical silent anonymity schemes (Mixzone, SLOW, RSP), the performance of TLAS is well-analyzed in terms of tractability rate and the average speed of nearby vehicles. Specifically,

TLAS dynamically configures the mix zone length based on the predicted traffic flow in the red light cycle, where the prediction algorithm is automatically selected by the scheme itself according to the multiple constrains. In the experiments, the coefficient γ_1 of CPU and γ_2 of memory are both 0.5; and the prediction algorithm running time is restricted to 3 seconds.

Figure 9 shows the performance comparison of the four schemes. As we can see, the security and anonymity of TLAS scheme and SLOW scheme are better than the other two schemes. Comparing with SLOW, TLAS can reduce the traceability rate and the standardized traceability rate by 1.8%

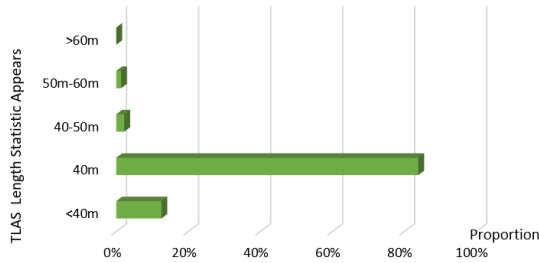


Fig. 10. Comparison of TLAS length statistic appears proportion.

and 8.2%, respectively. Besides, the mean total speed in TLAS is 0.07 m/s lower than in SLOW scheme, that means TLAS is more secure. Since SLOW scheme only considers the target vehicle's own speed, which ignores the speed of its neighbors. In addition, we also present the mix zone length distribution and prediction algorithm statistics of TLAS in Figure 10. The initial length of the silent mix zone (40m) meets the traffic flow demand in most cases. When the traffic flow is heavy, the mix zone length will increase accordingly. Due to the 50 vehicle limitation in the experiments, the largest mix zone length is 57.14m and only appeared a few times. In summary, TLAS scheme can achieve better performance in the aspect of anonymity and security.

VI. CONCLUSION

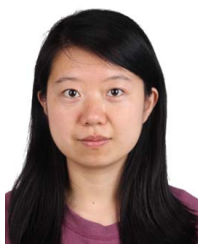
To protect the trajectory privacy in VANET, we propose a secure dynamic silent mix zone based pseudonym changing scheme (TLAS) based on the traffic condition for urban area. Considering the driving safety issues, the mix zone is defined in front of the red traffic light where vehicles are very slow or even stop. The mix zone length is dynamically configured according to the real-time traffic flow which is predicted in the green traffic light cycle. To well-balance the efficiency of anonymity and the resource utilization, four candidate traffic flow prediction algorithms are applied and one of them will be selected based on environment context. In addition, due to the mix zone length is calculated based on the prediction results, the error influence is also considered in terms of anonymous effect. Comparing with three typical silent anonymous schemes, comprehensive experimental results show that TLAS can achieve better performance in both anonymous effect and driving security.

In the future, we will analyze more aspects to improve TLAS, such as including a pseudonym exchange module to save the pseudonyms, leveraging multi-dimensional context data to design the zone, and using long-term prediction method to increase the accuracy. In addition, we also want to explore the mix context based strategies, as a supplement scheme for country area where road infrastructures are limited. Trajectory protection is an important issue, especially in modern intelligent transportation systems, adaptive and comprehensive schemes should be developed and widely applied to handle various environments.

REFERENCES

- [1] (Jul. 2020). *Connected Car Market by Service, Form, End Market, Network, Transponder, Hardware and Region—Global Forecast to 2025*. [Online]. Available: <https://www.marketsandmarkets.com/Market-Reports/connected-car-market-102580117.html>
- [2] C. Chen, Z. Liu, S. Wan, J. Luan, and Q. Pei, "Traffic flow prediction based on deep learning in internet of vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 6, pp. 3776–3789, Jun. 2020.
- [3] C. Chen, B. Liu, S. Wan, P. Qiao, and Q. Pei, "An edge traffic flow detection scheme based on deep learning in an intelligent transportation system," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 3, pp. 1840–1852, Mar. 2020.
- [4] S. Liu, L. Liu, J. Tang, B. Yu, and W. Shi, "Edge computing for autonomous driving: Opportunities and challenges," *Proc. IEEE*, vol. 107, no. 8, pp. 1697–1716, Jun. 2019.
- [5] C. Chen, Y. Zhang, Z. Wang, S. Wan, and Q. Pei, "Distributed computation offloading method based on deep reinforcement learning in ICV," *Appl. Soft Comput.*, vol. 103, May 2021, Art. no. 107108.
- [6] I. Memon, L. Chen, Q. A. Arain, H. Memon, and G. Chen, "Pseudonym changing strategy with multiple mix zones for trajectory privacy protection in road networks," *Int. J. Commun. Syst.*, vol. 31, no. 1, p. e3437, Jan. 2018.
- [7] J. Petit, F. Schaub, M. Feiri, and F. Kargl, "Pseudonym schemes in vehicular networks: A survey," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 1, pp. 228–255, Mar. 2014.
- [8] K. Emara, W. Woerndl, and J. Schlichter, "Context-based pseudonym changing scheme for vehicular adhoc networks," 2016, *arXiv:1607.07656*.
- [9] *IEEE Standard for Wireless Access in Vehicular Environments—Security Services for Applications and Management Messages*, IEEE Standard 1609.2-2016 (Revision IEEE Std 1609.2-2013), 2016, pp. 1–240.
- [10] *Intelligent Transport Systems (its); Security; Trust and Privacy Management*, document ETSI TS 102 941 V1.1.1, 2012.
- [11] A. Boulouache, S.-M. Senouci, and S. Moussaoui, "A survey on pseudonym changing strategies for vehicular ad-hoc networks," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 1, pp. 770–790, 1st Quart., 2017.
- [12] I. Saini, S. Saad, and A. Jaekel, "A comprehensive review of pseudonym changing strategies in vehicular networks," *IJ Netw. Secur.*, vol. 21, no. 5, pp. 785–796, 2019.
- [13] T. Gao and L. Zhao, "Pseudonym schemes based on location privacy protection in VANETs: A survey," in *Proc. Int. Conf. Innov. Mobile Internet Services Ubiquitous Comput. Cham, Switzerland: Springer*, 2020, pp. 597–605.
- [14] K. Emara, "Poster: PREXT: privacy extension for veins VANET simulator," in *Proc. IEEE Veh. Netw. Conf. (VNC)*, Dec. 2016, pp. 1–2.
- [15] M. Gerlach and F. Guttler, "Privacy in VANETs using changing pseudonyms—Ideal and real," in *Proc. IEEE 65th Veh. Technol. Conf. (VTC-Spring)*, Apr. 2007, pp. 2521–2525.
- [16] Y. Pan and J. Li, "An analysis of anonymity for cooperative pseudonym change scheme in one-dimensional VANETs," in *Proc. IEEE 16th Int. Conf. Comput. Supported Cooperat. Work Design (CSCWD)*, May 2012, pp. 251–257.
- [17] J. Liao and J. Li, "Effectively changing pseudonyms for privacy protection in VANETs," in *Proc. 10th Int. Symp. Pervasive Syst., Algorithms, Netw.*, 2009, pp. 648–652.
- [18] D. Eckhoff, C. Sommer, T. Gansen, R. German, and F. Dressler, "Strong and affordable location privacy in VANETs: Identity diffusion using time-slots and swapping," in *Proc. IEEE Veh. Netw. Conf.*, Dec. 2010, pp. 174–181.
- [19] D. Eckhoff, R. German, C. Sommer, F. Dressler, and T. Gansen, "SlotSwap: Strong and affordable location privacy in intelligent transportation systems," *IEEE Commun. Mag.*, vol. 49, no. 11, pp. 126–133, Nov. 2011.
- [20] A. Boulouache and S. Moussaoui, "TAPCS: Traffic-aware pseudonym changing strategy for VANETs," *Peer Peer Netw. Appl.*, vol. 10, no. 4, pp. 1008–1020, Jul. 2017.
- [21] J. Freudiger, M. Raya, and M. Felegghazi, "Mixzones for location privacy in vehicular networks," in *Proc. WiN-ITS*, Jan. 2007, pp. 1–7.
- [22] A. Boulouache and S. Moussaoui, "S2SI: A practical pseudonym changing strategy for location privacy in VANETs," in *Proc. Int. Conf. Adv. Netw. Distrib. Syst. Appl.*, Jun. 2014, pp. 70–75.
- [23] X. Li *et al.*, "PAPU: Pseudonym swap with provable unlinkability based on differential privacy in VANETs," *IEEE Internet Things J.*, vol. 7, no. 12, pp. 11789–11802, Dec. 2020.
- [24] A. Boulouache, S.-M. Senouci, and S. Moussaoui, "VLPZ: The vehicular location privacy zone," *Proc. Comput. Sci.*, vol. 83, pp. 369–376, Jan. 2016.
- [25] A. Boulouache, S.-M. Senouci, and S. Moussaoui, "Towards an efficient pseudonym management and changing scheme for vehicular ad-hoc networks," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2016, pp. 1–7.

- [26] R. Lu, X. Lin, T. Luan, X. Liang, and X. Shen, "Pseudonym changing at social spots: An effective strategy for location privacy in VANETs," *IEEE Trans. Veh. Technol.*, vol. 61, no. 1, pp. 86–96, Jan. 2011.
- [27] A. Boulouache and S. Moussaoui, "Urban pseudonym changing strategy for location privacy in VANETs," *Int. J. Ad Hoc Ubiquitous Comput.*, vol. 24, nos. 1–2, pp. 49–64, 2017.
- [28] B. Palanisamy and L. Liu, "MobiMix: Protecting location privacy with mix-zones over road networks," in *Proc. IEEE 27th Int. Conf. Data Eng.*, Apr. 2011, pp. 494–505.
- [29] A. Banihani, A. Alzahrani, R. Alharthi, H. Fu, and G. P. Corser, "T-PAAD: Trajectory privacy attack on autonomous driving," in *Proc. IEEE Conf. Commun. Netw. Secur. (CNS)*, May 2018, pp. 1–2.
- [30] Y. Cao, Y. Xiao, L. Xiong, and L. Bai, "PriSTE: From location privacy to spatiotemporal event privacy," in *Proc. IEEE 35th Int. Conf. Data Eng. (ICDE)*, Apr. 2019, pp. 1606–1609.
- [31] K. Emara, W. Woerndl, and J. Schlichter, "Vehicle tracking using vehicular network beacons," in *Proc. IEEE 14th Int. Symp. World Wireless, Mobile Multimedia Netw. (WoWMoM)*, Jun. 2013, pp. 1–6.
- [32] K. Emara, "Safety-aware location privacy in VANET: Evaluation and comparison," *IEEE Trans. Veh. Technol.*, vol. 66, no. 12, pp. 10718–10731, Dec. 2017.
- [33] (Jan. 1996). *Method of Measuring and Evaluating Congestion (GA115-1995)*. China Department of Traffic Management. [Online]. Available: <http://www.gb-gbt.com/PDF/Chinese.aspx/GA115-1995>
- [34] K. Haohui, C. Zhiying, B. Wenliang, and L. Hongru, "Using simple average ensemble modelling to the quantitative analysis of 4th analytes in cut tobacco samples," *Comput. Appl. Chem.*, vol. 5, no. 3, p. e9506, 2010.
- [35] R. J. Hyndman and Y. Khandakar, "Automatic time series forecasting: The forecast package for R," *J. Statist. Softw.*, vol. 27, no. 3, pp. 1–22, 2008.
- [36] S. J. Taylor and B. Letham, "Forecasting at scale," *Amer. Statistician*, vol. 72, no. 1, pp. 37–45, 2018.
- [37] L. Bieker, D. Krajewicz, A. Morra, C. Michelacci, and F. Cartolano, "Traffic simulation for all: A real world traffic scenario from the city of Bologna," in *Modeling Mobility With Open Data*. Cham, Switzerland: Springer, 2015.
- [38] P. A. Lopez *et al.*, "Microscopic traffic simulation using SUMO," in *Proc. 21st Int. Conf. Intell. Transp. Syst. (ITSC)*, Nov. 2018, pp. 2575–2582.
- [39] K. Sampigethaya, L. Huang, M. Li, R. Poovendran, K. Matsuura, and K. Sezaki, "CARAVAN: Providing location privacy for VANET," in *Proc. Workshop Embedded Secur. Cars (ESCAR)*, 2005, pp. 1–15.
- [40] L. Buttyan, T. Holczer, A. Weimerskirch, and W. Whyte, "SLOW: A practical pseudonym changing scheme for location privacy in VANETs," in *Proc. IEEE Veh. Netw. Conf. (VNC)*, Oct. 2009, pp. 1–8.
- [41] L. Buttyán, T. Holczer, and I. Vajda, "On the effectiveness of changing pseudonyms to provide location privacy in VANETs," in *Proc. Eur. Workshop Secur. Ad-Hoc Sensor Netw.* Berlin, Germany: Springer, 2007, pp. 129–141.



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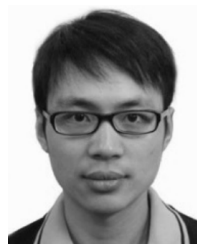
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